

1. Title

Practical 2: Teaching Ratings Data Analysis Using Python

2. Problem Statement

The objective of this practical is to analyze the Teaching Ratings dataset containing information about professors and their teaching evaluations. The dataset includes variables such as age, gender, beauty score, evaluation score, tenure status, and number of students.

The following tasks are performed:

1. Determine whether the teaching ratings data is time-series or cross-sectional.
2. Calculate the mean, median, minimum, and maximum number of students.
3. Generate the descriptive statistics of the numerical variables.
4. Create a histogram of the beauty variable and interpret its distribution.
5. Compare the average beauty score by gender using mean and standard deviation.
6. Calculate the percentage of male and female instructors who are tenured and determine whether tenure status differs by gender.

3. Solution

Here is the complete python code with outputs.

4. Conclusion

The Teaching Ratings dataset is **cross-sectional** and contains 458 observations. The average number of students is approximately **36.64**, while the median is **23**.

The beauty scores are slightly positively skewed. Female instructors have a higher average beauty score than male instructors. In terms of tenure, **80.99% of male instructors** and **74.36% of female instructors** are tenured, indicating a difference in tenure proportions between the two genders.

Import Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

Load Dataset

```
df = pd.read_csv("/content/teachingratings_extracted.csv")

df.head()
```

	minority	age	gender	credits	beauty	eval	division	native	tenure	students	allstudents	prof	PrimaryLast	vismin	f
0	yes	36	female	more	0.289916	4.3	upper	yes	yes	24	43	1	0	1	
1	yes	36	female	more	0.289916	3.7	upper	yes	yes	86	125	1	0	1	
2	yes	36	female	more	0.289916	3.6	upper	yes	yes	76	125	1	0	1	
3	yes	36	female	more	0.289916	4.4	upper	yes	yes	77	123	1	1	1	
4	no	59	male	more	-0.737732	4.5	upper	yes	yes	17	20	2	0	0	

Next steps:

[Generate code with df](#)
[New interactive sheet](#)

Q1. Is the teachers' rating data time series or cross-sectional?

```
# Display column names
print(df.columns.tolist())

# Check whether there is a date/time variable
```

```
print("\nData types:")  
print(df.dtypes)
```

```
['minority', 'age', 'gender', 'credits', 'beauty', 'eval', 'division', 'native', 'tenure', 'students', 'allstudents', 'prof',
```

```
Data types:
```

```
minority      object  
age           int64  
gender        object  
credits        object  
beauty        float64  
eval          float64  
division       object  
native         object  
tenure         object  
students       int64  
allstudents    int64  
prof           int64  
PrimaryLast    int64  
vismin         int64  
female         int64  
single_credit  int64  
upper_division int64  
English_speaker int64  
tenured_prof   int64  
dtype: object
```

It represents observations of professors/courses rather than observations recorded sequentially over time. Although the same professor can appear in multiple rows because they have multiple course evaluations, there is no time dimension.

Q2. Find the mean, median, minimum and maximum values for students

```
mean_students = df['students'].mean()  
median_students = df['students'].median()  
min_students = df['students'].min()  
max_students = df['students'].max()  
  
print("Mean:", mean_students)
```

```
print("Median:", median_students)
print("Minimum:", min_students)
print("Maximum:", max_students)
```

```
Mean: 36.64192139737991
Median: 23.0
Minimum: 5
Maximum: 380
```

Q3. Produce the descriptive statistics table

```
# Descriptive statistics for numerical variables
df.describe()
```

	age	beauty	eval	students	allstudents	prof	PrimaryLast	vismin	female	single_credit
count	458.000000	458.000000	458.000000	458.000000	458.000000	458.000000	458.000000	458.000000	458.000000	458.000000
mean	48.323144	-0.001213	3.998035	36.641921	55.268559	45.379913	0.200873	0.139738	0.425764	0.058952
std	9.816052	0.787657	0.552948	45.125434	75.364874	27.529491	0.401092	0.347094	0.494999	0.235792
min	29.000000	-1.450494	2.100000	5.000000	8.000000	1.000000	0.000000	0.000000	0.000000	0.000000
25%	42.000000	-0.656269	3.600000	15.000000	19.000000	20.000000	0.000000	0.000000	0.000000	0.000000
50%	48.000000	-0.068014	4.000000	23.000000	29.000000	44.000000	0.000000	0.000000	0.000000	0.000000
75%	57.000000	0.547945	4.400000	40.000000	60.000000	70.750000	0.000000	0.000000	1.000000	0.000000
max	73.000000	1.970023	5.000000	380.000000	581.000000	94.000000	1.000000	1.000000	1.000000	1.000000

```
descriptive_stats = df.describe().T
```

```
descriptive_stats
```

	count	mean	std	min	25%	50%	75%	max
age	458.0	48.323144	9.816052	29.000000	42.000000	48.000000	57.000000	73.000000
beauty	458.0	-0.001213	0.787657	-1.450494	-0.656269	-0.068014	0.547945	1.970023
eval	458.0	3.998035	0.552948	2.100000	3.600000	4.000000	4.400000	5.000000
students	458.0	36.641921	45.125434	5.000000	15.000000	23.000000	40.000000	380.000000
allstudents	458.0	55.268559	75.364874	8.000000	19.000000	29.000000	60.000000	581.000000
prof	458.0	45.379913	27.529491	1.000000	20.000000	44.000000	70.750000	94.000000
PrimaryLast	458.0	0.200873	0.401092	0.000000	0.000000	0.000000	0.000000	1.000000
vismin	458.0	0.139738	0.347094	0.000000	0.000000	0.000000	0.000000	1.000000
female	458.0	0.425764	0.494999	0.000000	0.000000	0.000000	1.000000	1.000000
single_credit	458.0	0.058952	0.235792	0.000000	0.000000	0.000000	0.000000	1.000000
upper_division	458.0	0.661572	0.473692	0.000000	0.000000	1.000000	1.000000	1.000000
English_speaker	458.0	0.938865	0.239840	0.000000	1.000000	1.000000	1.000000	1.000000
tenured_prof	458.0	0.781659	0.413572	0.000000	1.000000	1.000000	1.000000	1.000000



Next steps:

[Generate code with descriptive_stats](#)
[New interactive sheet](#)

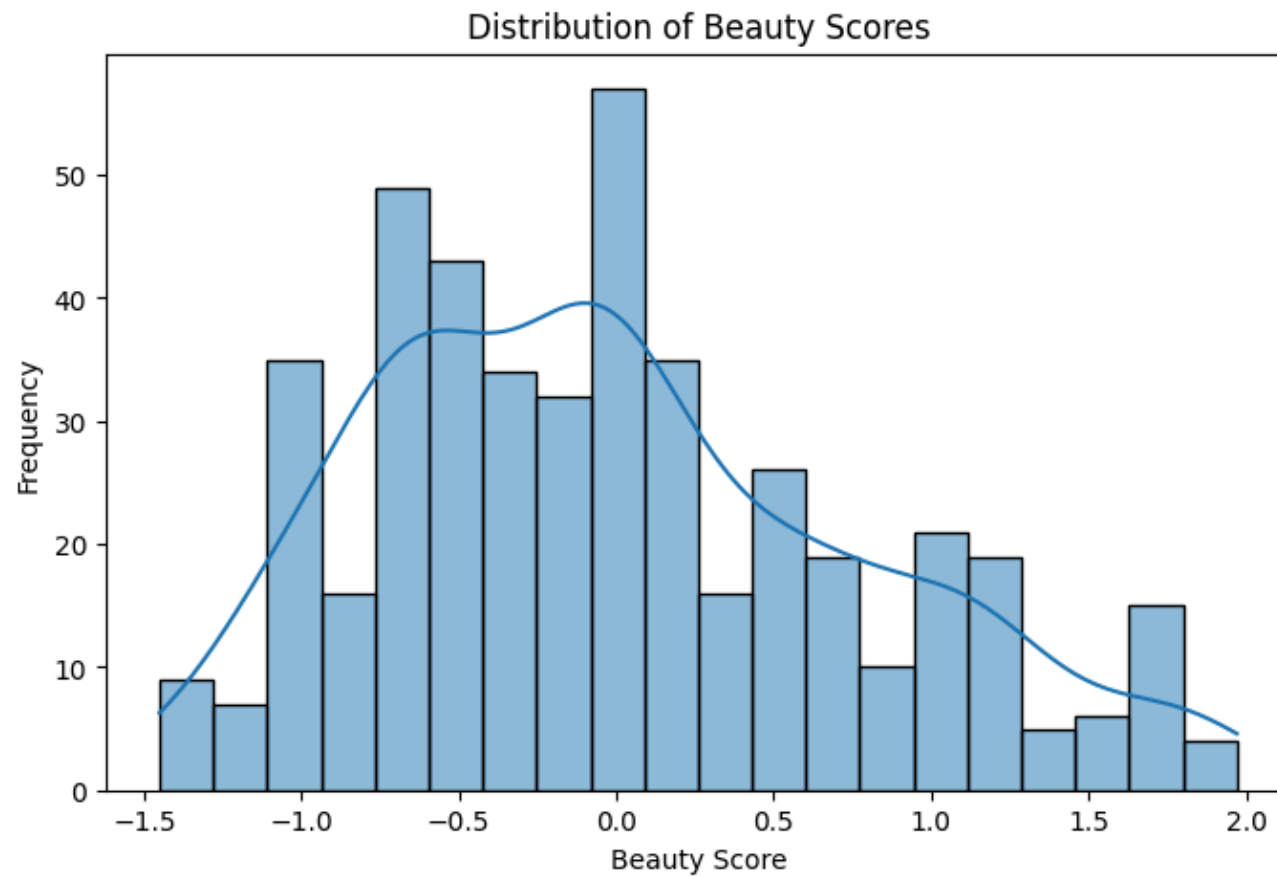
Q4. Create a histogram of the beauty variable

```
plt.figure(figsize=(8,5))

sns.histplot(df['beauty'], bins=20, kde=True)

plt.title('Distribution of Beauty Scores')
plt.xlabel('Beauty Score')
plt.ylabel('Frequency')

plt.show()
```



```
print("Skewness:", df['beauty'].skew())
```

```
Skewness: 0.5047636965025605
```

The beauty scores are centered close to 0. The distribution is slightly positively/right skewed, meaning there is a somewhat longer tail toward higher beauty scores. It is not extremely skewed, so the distribution is reasonably balanced overall.

Q5. Does average beauty score differ by gender?

```
gender_beauty = df.groupby('gender')['beauty'].agg(
    Mean='mean',
    Standard_Deviation='std',
    Count='count'
)

gender_beauty.round(3)
```

	Mean	Standard_Deviation	Count	
gender				
female	0.116	0.818	195	
male	-0.088	0.754	263	

Female instructors have an average beauty score of approximately 0.116, whereas male instructors have an average beauty score of approximately -0.088.

Therefore, yes, the average beauty score differs by gender in this dataset. Female instructors have a somewhat higher average beauty score than male instructors

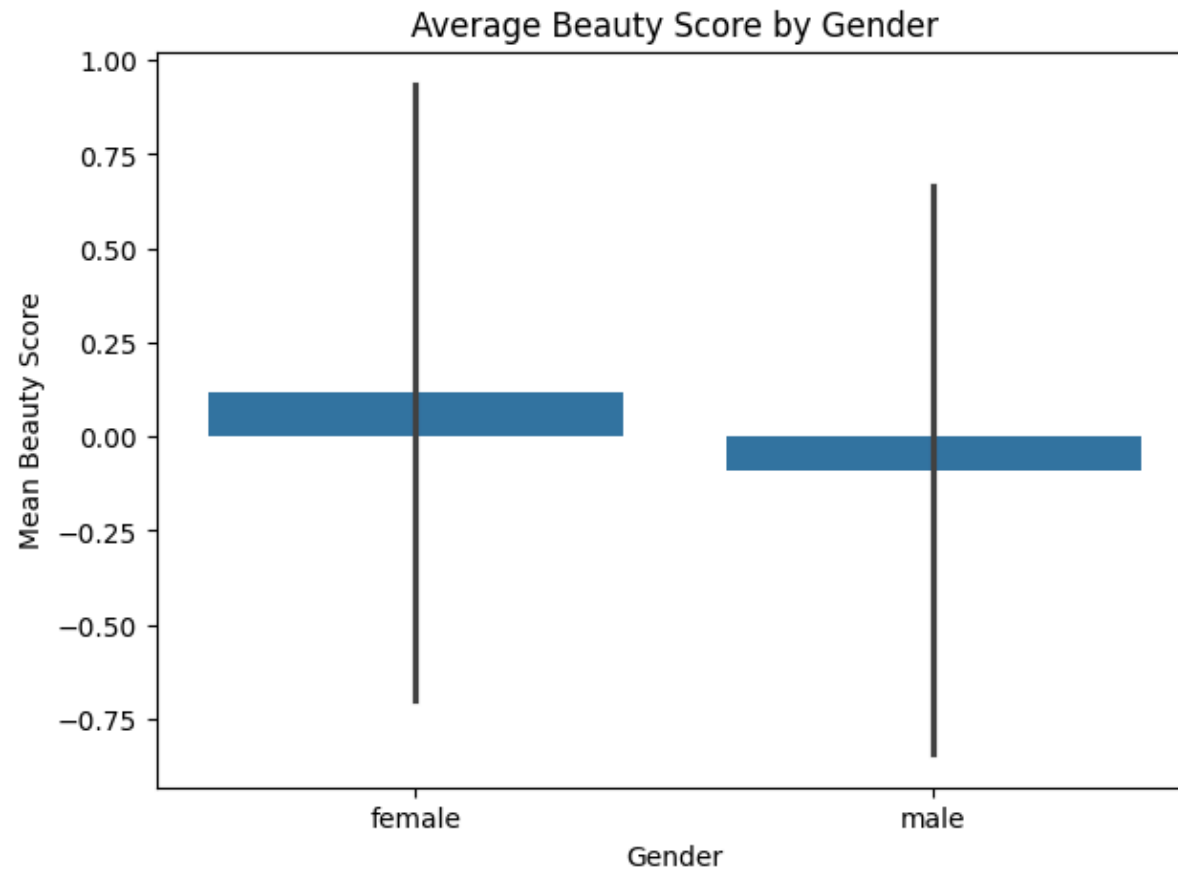
Optional visualization

```
plt.figure(figsize=(7,5))

sns.barplot(
    data=df,
    x='gender',
```



```
y='beauty',  
errorbar='sd'  
)  
  
plt.title('Average Beauty Score by Gender')  
plt.xlabel('Gender')  
plt.ylabel('Mean Beauty Score')  
  
plt.show()
```



Q6. Calculate the percentage of males and females that are tenured professors

```
print(df['tenure'].value_counts())
```

```
tenure
yes    358
no     100
Name: count, dtype: int64
```

```
tenure_gender = pd.crosstab(
    df['gender'],
    df['tenure'],
    normalize='index'
) * 100
```

```
tenure_gender.round(2)
```

tenure	no	yes
female	25.64	74.36
male	19.01	80.99

Interpretation

Approximately:

74.36% of female instructors are tenured 80.99% of male instructors are tenured

Therefore, the percentage of tenured professors does differ by gender in this dataset. The proportion of tenured male instructors is higher than the proportion of tenured female instructors.

Start coding or [generate](#) with AI.